MRDVS
Mobile Robot Vision Expert



V2 Pro Localization Module

Product White Paper

Multi-sensor fusion localization for complex environments

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1 Product Introduction

V2 Pro integrates upward-looking vision, 2D LiDAR, IMU and onboard computing to map the environment and provide real-time position, heading and localization status.

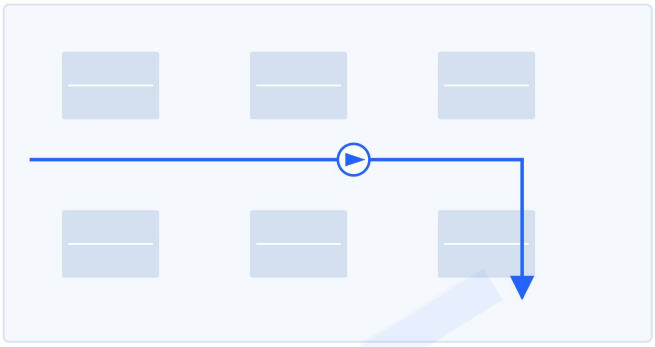
V2 Pro

Localization module



Mounted on a mobile platform

Locates the vehicle on the map



Position X, Y Heading yaw

Key outputs

Planar position

Vehicle X, Y coordinates in the current map

Heading

Vehicle orientation relative to the map

Localization status

Indicates whether localization is valid

From installation to localization

1 Mount and power

- Mount the module level
- Keep sensor views clear
- Connect 24 V power

2 Configure

- Set up the network
- Set mounting extrinsics
- Position and orientation

3 Build the map

- Capture operating routes
- Generate and check map
- Load the required map

4 Localize

- Set the initial position
- Check position and heading
- Output localization data

Mounting extrinsics describe the module's position, orientation and height relative to the vehicle.

▶ Watch the Live Demo on YouTube

1.1 Development History

A decade of evolution, from upward-looking vision to sensor fusion

Robotics experience
Upward-looking localization for mobile platforms

▶

Multi-sensor fusion
Vision, LiDAR and IMU for vehicle localization

▶

V2 Pro integration
Sensors and processing in a single module

Origins: upward-looking visual localization

V2 Pro traces its origins to the localization system used in the early autonomous platforms of Lanxin Robotics. An upward-looking infrared camera, combined with deep-learning-enhanced visual simultaneous localization and mapping (VSLAM), formed an in-house positioning engine for mobile platforms.

Field validation and continued development

The technology was deployed across thousands of mobile platforms, gaining field experience at customers in China's 3C electronics, packaging, photovoltaic and automotive industries. In 2023, MRDVS built on this accumulated expertise to focus further on spatial intelligence and safety performance for mobile robots.

Fusion: bringing LiDAR, vision and IMU together

Building on the visual localization foundation, the development team integrated a 2D LiDAR with the visual-inertial module. The architecture evolved from co-located sensors toward tighter coupling. V2 Pro unifies LiDAR, vision and IMU at both the hardware and algorithm levels in a single module calibrated at the factory.

Productization: extending proven robotics experience

The positioning core originally used in Lanxin mobile platforms became the self-contained MRDVS V2 Pro product. Equipment manufacturers and system integrators can apply it to forklift retrofits and new mobile equipment designs, while using real-time pose data to support safety applications for mobile equipment and digital twin systems.

1.2 Key Advantages

01 Centimeter-level accuracy and real-time output

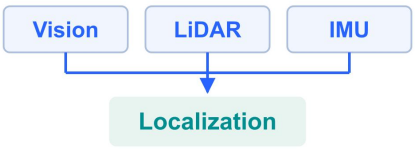
1–3 cm	20 Hz	<20 ms
Localization error	Valid pose output rate	Output latency

02 Built for industrial localization

Upward-looking vision, LiDAR and IMU fusion

Sensors and algorithms designed for industrial use

No unnecessary sensors or associated cost.



03 Large-area mapping and localization

Per mapping session: 20,000 m² and 2,000 m path

Extend maps incrementally to cover large sites.

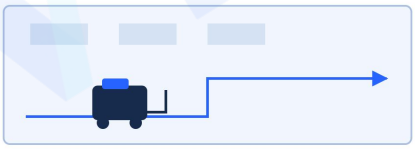


04 No positioning base stations

Compared with typical UWB anchor systems,

avoid anchor installation, cabling and maintenance.

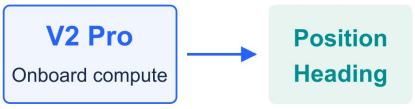
Natural features; QR codes can assist recovery.



05 No external localization computer

Camera, LiDAR, IMU and processing in one module

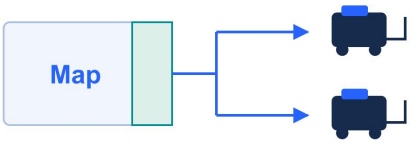
Mapping and localization run onboard V2 Pro.



06 Map expansion and reuse across vehicles

Extend maps into new areas and reuse them on other vehicles

Expand localization coverage as operations grow.

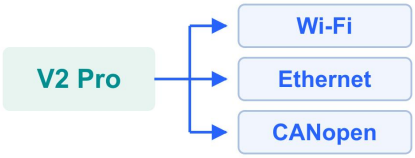


07 Wi-Fi and multiple communication options

Wi-Fi, Ethernet and CAN (CANopen) connectivity.

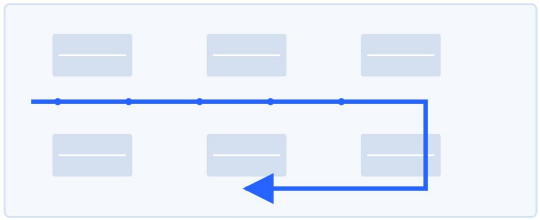
Integrate via UDP, HTTP, MQTT, ROS 1 / ROS 2.

Connect to the platform and existing control systems.



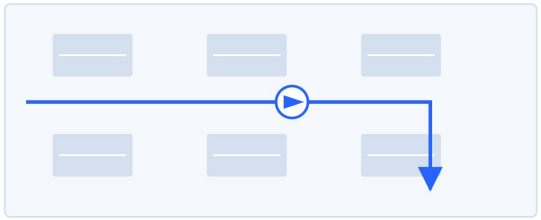
1.3 Main Functions

Create maps



Capture the operating environment
Build a map for localization

Continuous localization



Localize within the mapped area
Output vehicle position and heading

Relocalization



Use a QR code or enter a pose
Re-establish vehicle position

Map management



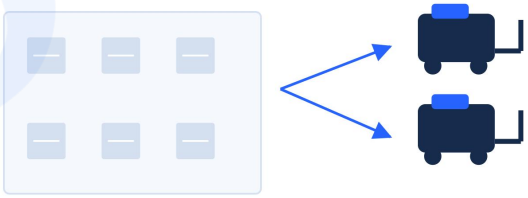
View and load maps on the device
Export or delete maps

Incremental mapping



Capture missed areas
Extend maps into new areas

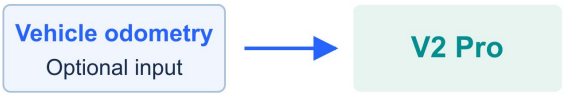
Map reuse



Use a map on other vehicles
Subject to mounting and setup requirements

Vehicle odometry assistance

Accepts wheel speed or vehicle odometry
Helps improve accuracy and localization success.

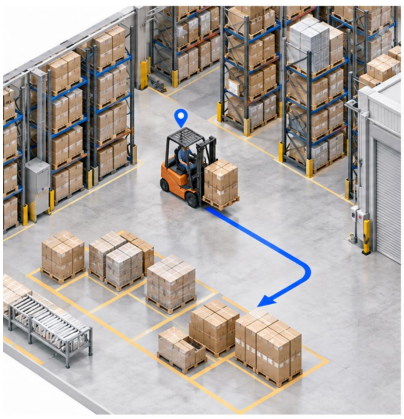


2 Applications

Position and heading for mobile equipment in warehousing, production and intralogistics.

Warehouse handling

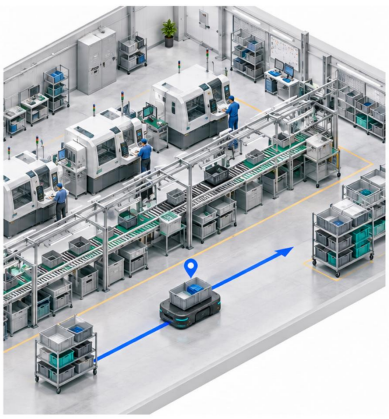
Aisles and receiving / dispatch



Inbound, outbound and internal pallet moves

Production supply

Line-side storage and workstations



Material supply and workstation transfers

Intralogistics

Warehouses, workshops and links



Transport between operating areas





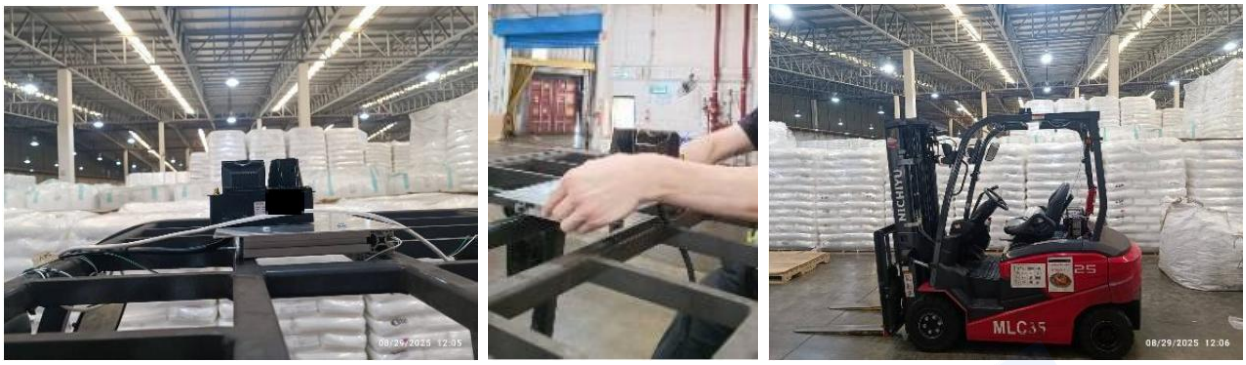


Illustrative scenes

2.1 Project Examples

2.1.1 Yusen Logistics Warehouse Validation

At Yusen Logistics' WH4 warehouse, V2 Pro was fitted to a forklift to evaluate localization while driving, turning and carrying loads.



Site	Vehicle	Validation scope
Yusen Logistics WH4 warehouse	**Forklift**	**Mapping and continuous localization**

Customer needs

1. Locate the forklift in real time
2. Determine its heading
3. Maintain localization while handling loads

V2 Pro solution



V2 Pro

- Camera detects overhead features
- LiDAR scans surrounding structures
- Module computes vehicle position

Position + heading

On-site validation

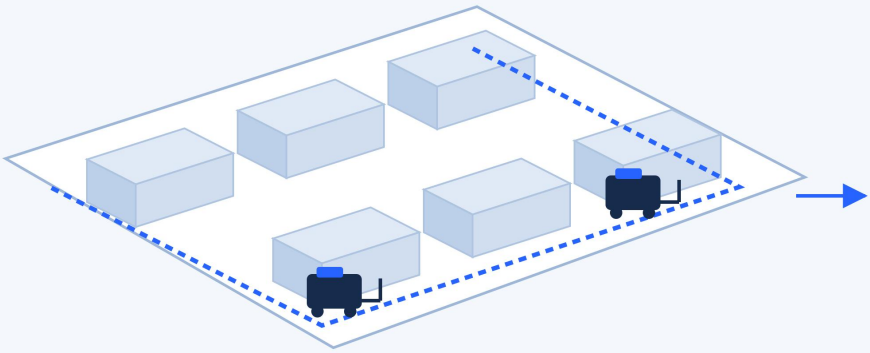
1. **Map the warehouse**
   Capture the operating area
   Generate a localization map
2. **Test driving and turning**
   Check position and heading during driving and turns
3. **Test loaded operation**
   Check localization continuity while carrying loads

2.1.2 Manufacturing Site Deployment

A compressor manufacturer needed to locate a small fleet of forklifts across a 5,000 m² site while reducing investment in positioning infrastructure.

5,000 m²
Factory site · Forklifts

Deployment concept



Localization data
- Planar position X / Y
- Vehicle heading
- Localization status

One module per vehicle · Natural environmental features · No positioning base stations

Customer needs
- View forklift positions across the site
- Update positions as vehicles move
- Reduce positioning infrastructure costs

V2 Pro deployment
- Mount a localization module on each forklift
- Compute position and heading onboard
- Build a map and send poses to the customer system

Project benefits

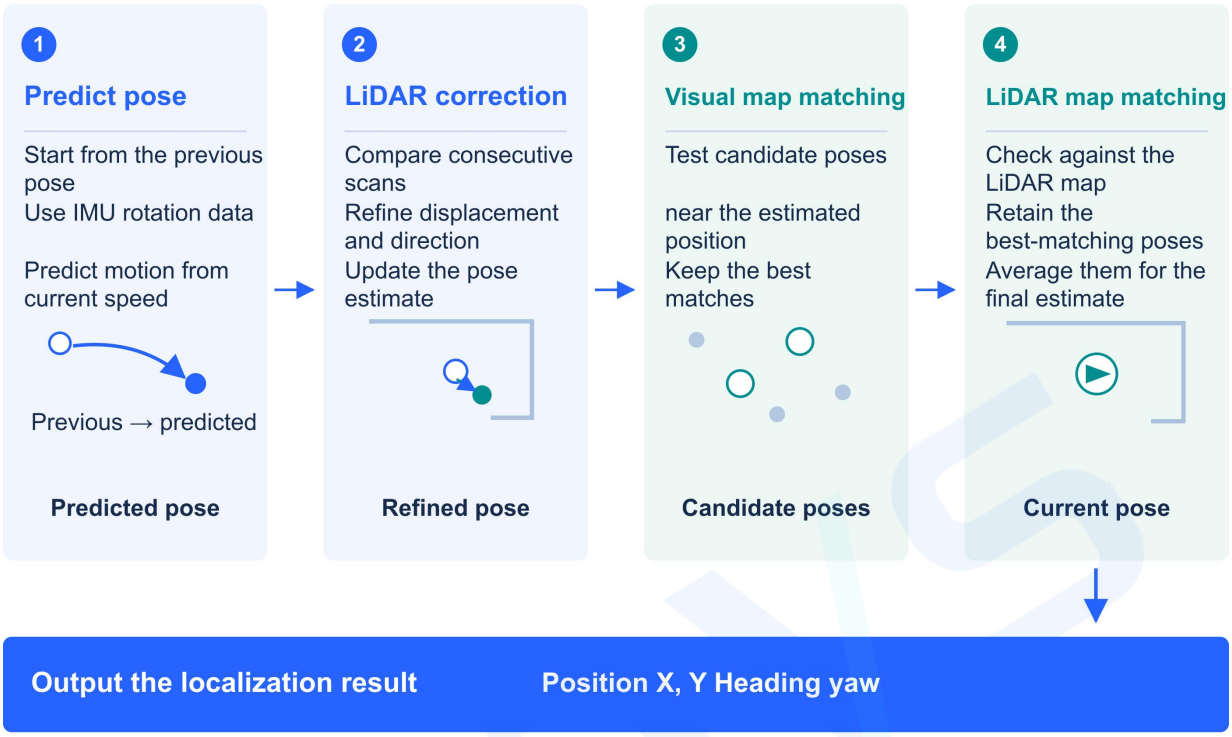
Vehicle visibility
Position and heading output
Shows each forklift's current location

Less site infrastructure
Vehicle-mounted localization
No positioning base stations on site

System integration
Vehicle position data
Supports route recording in customer systems

3 Localization Principle

V2 Pro predicts vehicle motion, then refines the pose using LiDAR scans, the visual map and the LiDAR map.



Why the map matters A fixed reference helps correct drift that accumulates in continuous motion estimates.

Repeat this process as new sensor data arrives.

4 Localization Performance

> **These specifications apply when installation, environmental and map requirements are met. They cover accuracy, motion, output timing and relocalization.**

Localization Accuracy

1–3 cm	**±1°**
Localization error	Heading error

Motion Limits

15 km/h	**3 km/h**	**Approx. 28.6°/s**
Max. localization speed	Max. mapping speed	Max. yaw rate

Output and Startup

20 Hz	**<20 ms**	**>99.9%**	**40 s**
Valid pose output rate	Output latency	Localization availability	Power-on to localization

Relocalization

4 s	**99%**	**<8 km/h**	**10 m**
Relocalization time	Relocalization success rate	Speed through QR detection area	QR detection distance

Understanding the Specifications

Output rate and latency: A valid pose is a position and heading with valid localization status. At 20 Hz, updates are about 50 ms apart. The update interval and output latency are different measures.

Availability and relocalization success: Availability describes the usability of ongoing localization. Relocalization success is the proportion of successful attempts to re-establish localization.

Startup and relocalization: 40 s is the time from power-on to localization; 4 s is the time to re-establish localization.

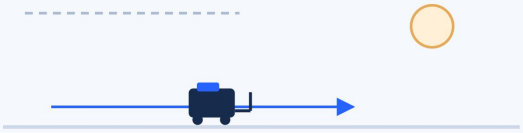
Yaw rate: Approx. 28.6°/s is the maximum change in heading per second. Wheel slip may cause a mismatch between physical and LiDAR odometry, resulting in localization drift.

QR relocalization: <8 km/h is the vehicle speed through the QR detection area; 10 m is the detection distance. The QR code must be registered in the current map.

5 System Limits

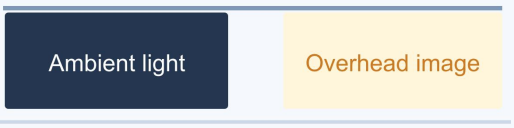
5.1 Environmental Requirements

V2 Pro is primarily for indoor use. Limited, brief outdoor sections require assessment on the actual route.



Brief outdoor sections

Limited, short outdoor sections are supported.
Assess the actual operating route.
Not for prolonged continuous outdoor use.



Lighting and overhead features

Ambient light: 0–10 klux.
Artificial light has no impact; overhead texture is needed.
Prolonged direct sunlight can disable visual localization.



Features and extraction range

Continuous featureless path: ≤20 m
Maximum LiDAR feature extraction range: 40 m
Assess sparse features or repetitive structures.
Lower reflectivity reduces effective range.



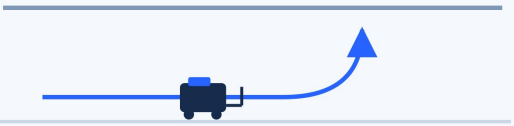
Unobstructed sensor views

Keep both the camera and LiDAR fields of view clear.



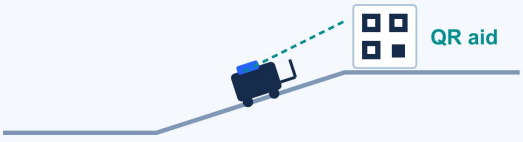
Dynamic environments

Supports dynamic sites and major structural changes.
LiDAR odometry helps accommodate environmental changes.
Small shifts across large structural areas need attention.



Turning and vehicle motion

Maximum yaw rate: approx. 28.6°/s
Wheel slip may cause odometry mismatch and shift the estimated position.
Localization ≤15 km/h; mapping ≤3 km/h



Slopes and uneven floors

Slopes may cause localization loss; use QR assistance.
Assess sites with frequent slopes or large height changes.



Vibration and bumps

Mount the module securely.
For significant vibration or bumps, validate localization performance on site.

5.2 Mapping and Map Use

Keep each mapping session within the distance and area limits, and avoid long, narrow areas. Incremental mapping must overlap the existing base map. Map reuse also depends on mounting conditions.

Path length per mapping session	Area per mapping session	Area added per incremental session
2,000 m	20,000 m²	5,000 m²

Total area per map, including extensions: 250,000 m²

This limit primarily reflects the software platform's map-rendering capacity. Customers using their own interface to send relocalization positions and monitor localization are not subject to this interface limit.

2,000 m is the cumulative path length per mapping session, not the maximum length of a site.

Site shape and incremental mapping

Avoid long, narrow, feature-poor areas



Long, narrow areas with few features can distort the map.

Assess site length and aspect ratio against the route.

Same shape requirements for new maps and extensions.

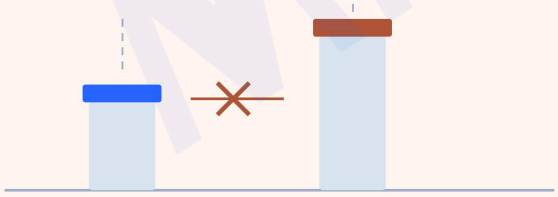
New coverage must overlap the base map



Start and finish in the base map.
Add overlap points along the route.

2D maps and mounting height

No direct reuse at a different height



The same map cannot be used directly at a different mounting height.

Small height changes must meet both conditions



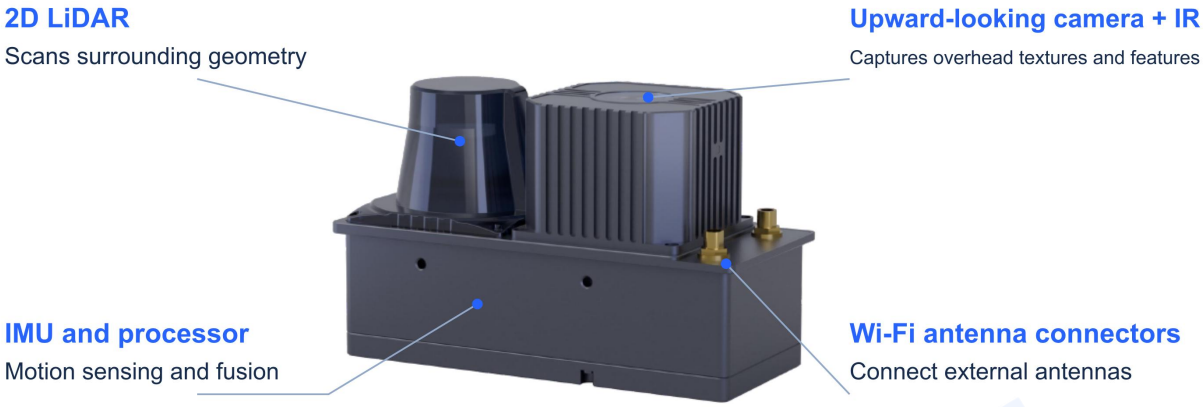
Surrounding LiDAR features unchanged

Validate the actual operating routes after reuse.

Localization uses a 2D map and outputs X, Y and heading.

6 Hardware and Installation

The camera, 2D LiDAR, IMU and processor are integrated in one module.



146.5 × 66.3 × 89.4 mm
Overall dimensions

6.6 W / 15.3 W
Average / maximum power

24 V DC / 3 A
Power supply

Hardware	Specification
Upward-looking camera	Working distance: 1–15 m to overhead features; 20 fps Field of view: 108° (H) × 57° (V), ±3° Ambient light: 0–10 klux Glare and ambient-light resistance Built-in infrared illumination
2D LiDAR	270° scan angle; selectable 15–25 Hz Range: 15 m at 10% / 40 m at 90% reflectivity
LiDAR measurement	Typical ranging accuracy: ±30 mm; angular resolution: 0.25°
IMU and processing	Built-in IMU; 8-core ARM processor; 6 TOPS NPU
Dimensions	146.5 × 66.3 × 89.4 mm
Weight	Approx. 543 g
Memory and storage	4 GB RAM; 64 GB storage

Certification and compliance

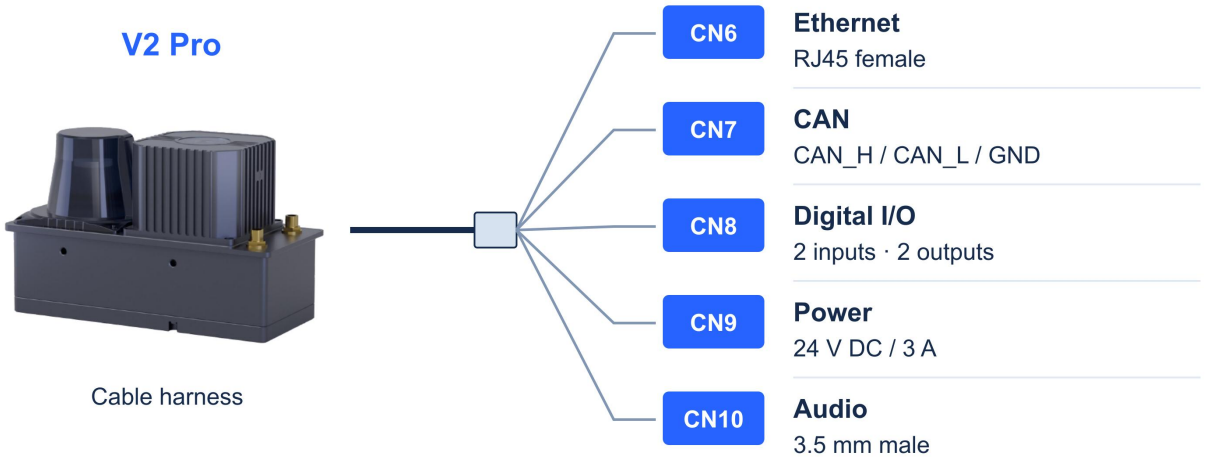
EMC — CE-EMC
Emissions and immunity tests passed

Radio and electrical safety — RED
RF, health and electrical safety tests completed

Laser safety — EN 60825
Laser safety testing completed

Substance compliance — RoHS · REACH
Restricted substances and chemical compliance

6.1 Interfaces and Electrical Specifications



Interface	Specification
CN6 · Ethernet	RJ45; 100 Mbit/s Ethernet
CN7 · CAN	CANopen; default 250 kbit/s, adjustable
CN8 · Digital I/O	2 inputs: 5–24 V, active high 2 outputs: push-pull, 24 V / 100 mA
CN9 · Power	24 V DC / 3 A; input range: 19.2–28.8 V
CN10 · Audio	3.5 mm audio output
Wi-Fi	2.4 GHz; 2 antennas

Network interfaces: HTTP, MQTT, UDP, ROS 1 / ROS 2.

Operating conditions

Parameter	Specification
Power consumption	6.6 W average; 15.3 W maximum
Operating temperature	−20 to 60 °C
Storage temperature	−40 to 80 °C
Ingress protection	IP54
Power source	Continuous external power required
Storage humidity	≤95% RH, non-condensing
ESD immunity	±4 kV contact; ±8 kV air discharge (tested)

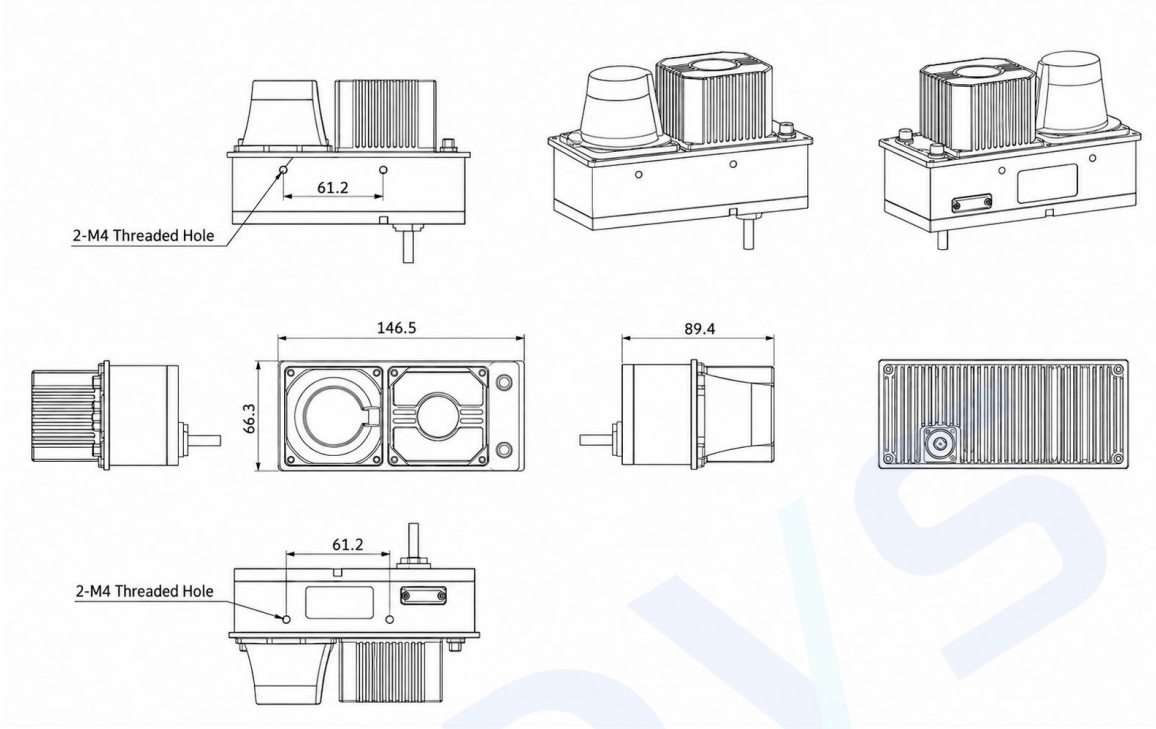
RTC Backup Battery

Parameter	Specification
Purpose	Maintains the real-time clock when main power is off; does not power localization
Electrical ratings	Rechargeable; 3.0 V nominal; 5.8 mAh standard capacity
Size and weight	6.8 mm diameter × 2.1 mm height; approx. 250 mg
Installation and service	Soldered in place; not user-removable or user-replaceable

6.2 Dimensions and Mounting Location

146.5 × 66.3 × 89.4 mm

Length × width × height. Drawing dimensions in mm.



Forklift roof mounting example



Example: manned forklift

Mounting location

In this example, the LiDAR faces rearward to avoid mast obstruction. Mount on a rigid roof structure and keep the scan clear.

Clearance

Allow room for downward cable routing, cooling, cleaning and connector access.

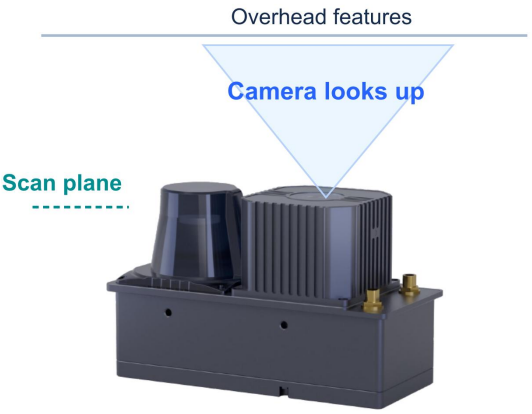
Installation drawing assistance

We can help prepare an installation drawing based on the vehicle structure and mounting location.

6.3 Mounting Requirements and Procedure

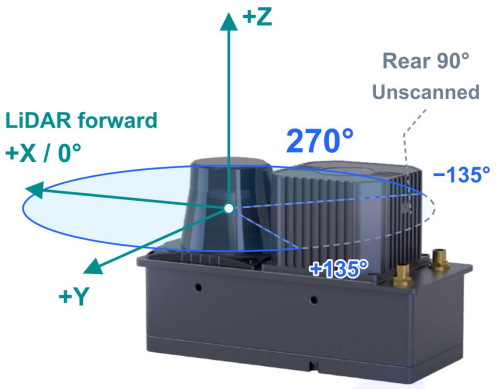
Mounting requirements

Sensor views



Level mounting; keep sensor views clear.
Camera up; cable exits downward.

LiDAR field of view and axes



+X forward · +Y left · +Z up
Scan: −135° to +135° relative to +X

Installation guide

Mount level



Base cable exit



01 Select the location: Use a rigid vehicle structure. Keep the camera and LiDAR fields of view clear.

02 Mount level: Keep the module horizontal, with the LiDAR facing forward or backward along the direction of travel.

03 Connect the cables: Route cables downward from the base. Secure them and allow sufficient bend clearance.

04 Configure and verify: Set up the network and mounting extrinsics. Power on, check operation and validate the actual routes.

7 Software Use

7.1 Platform Connection and Overview

One interface for deployment, mapping, relocalization and localization monitoring.



01 Deployment setup

Configure the network, mounting extrinsics and device settings.

02 Mapping and map management

Create maps, view mapping progress and manage saved maps.

03 Relocalization

Set the vehicle's position and heading on the map.

04 Localization monitoring

View vehicle position, heading and localization status.

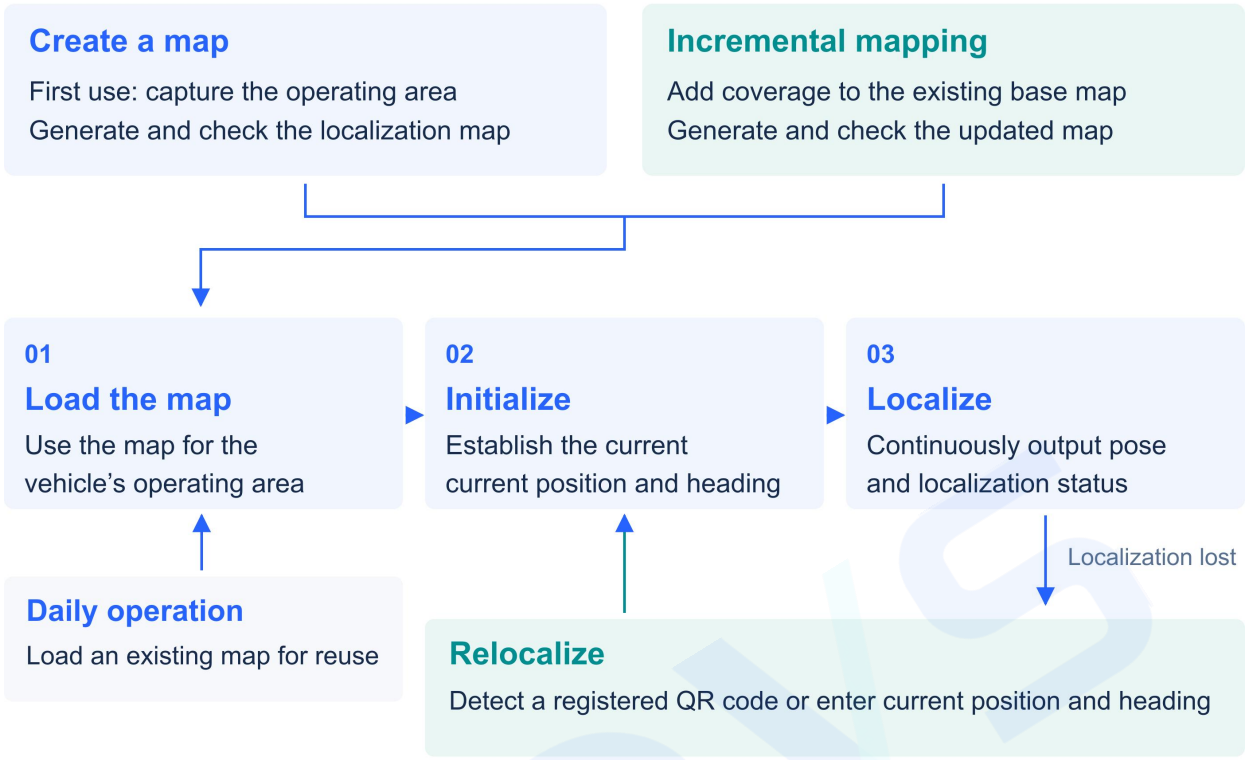
How the platform and V2 Pro work together

Software platform		V2 Pro module
Sends settings and commands	⟷	Captures the environment
Displays maps and localization results		Computes maps and vehicle pose

Customer systems can use position data for zone management, route analysis and operating statistics.

7.2 Mapping and Localization

Prepare a map



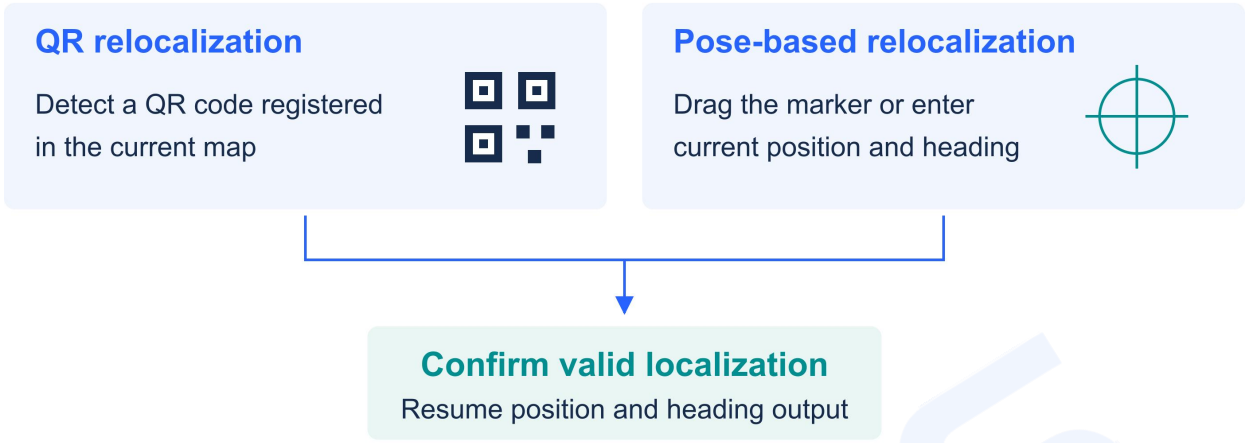
A shared spatial reference

Outputs 2D position X, Y and heading within the mapped area.

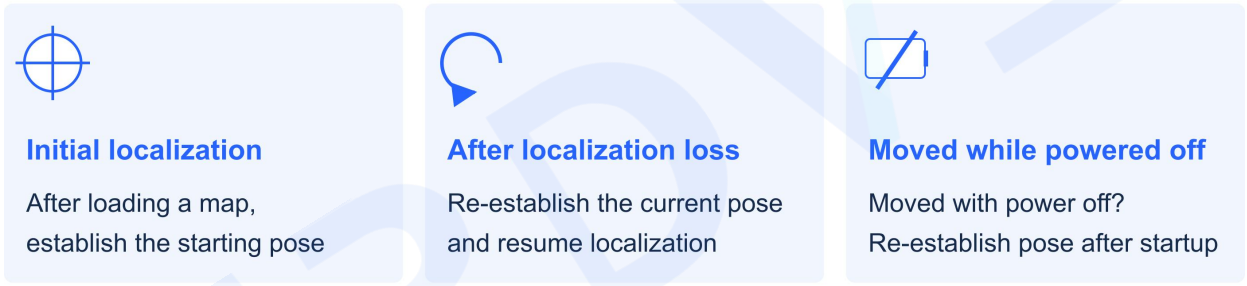
7.3 Relocalization

Registered QR codes, dragging the vehicle marker, or entering the current position and heading can re-establish localization.

Two ways to re-establish the current pose



When to use it



Before relocalizing, confirm that the loaded map matches the vehicle's operating area.

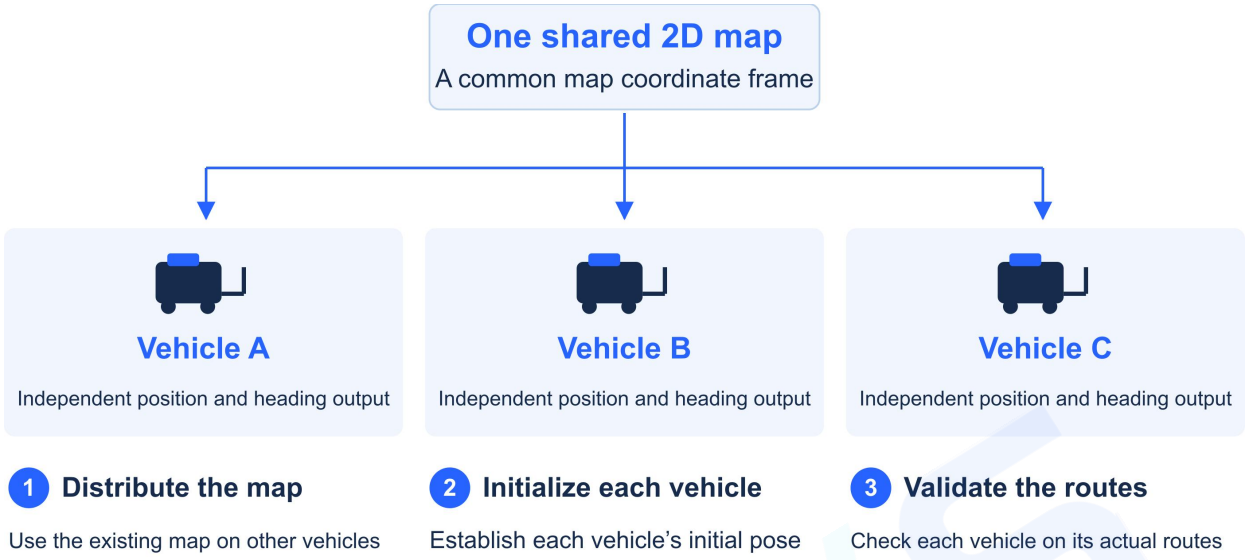
When QR codes are needed

In typical environments, V2 Pro localizes using natural features without dedicated QR codes. Install QR codes to assist localization in unusual environments, areas with few distinctive features, or locations where slopes affect localization.

QR codes must be registered in the current map and within the camera's detection range.

7.4 Map Reuse Across Vehicles

An existing map can be used on other vehicles. Each vehicle initializes localization and outputs its own pose. Map each floor separately and use the map for the current floor.



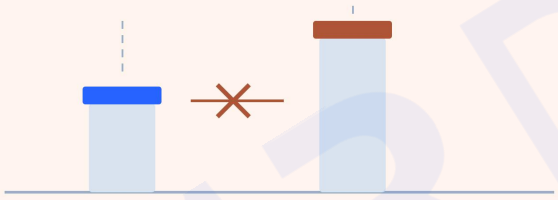
1 Distribute the map
Use the existing map on other vehicles

2 Initialize each vehicle
Establish each vehicle's initial pose

3 Validate the routes
Check each vehicle on its actual routes

Mounting height and map reuse

No direct reuse at a different height



The same map cannot be used directly at a different mounting height.

Small height changes must meet both conditions



Surrounding LiDAR features unchanged

Validate the actual operating routes after reuse.

Localization uses a 2D map and outputs X, Y and heading.

8 Communications and Data Integration

V2 Pro exchanges localization, map and device information with vehicle controllers or management systems.

8.1 Connections and Protocols

Connection	Protocols and data interfaces
Ethernet / Wi-Fi	UDP · HTTP · MQTT · ROS 1 / ROS 2
CAN	CANopen · 250 kbit/s default (adjustable)

8.2 Inputs and Outputs



Input fields	Output fields
Mapping and map control Control command, map name	**Localization** Position, heading, pose timestamp, localization status
Relocalization Initial position, initial heading	**Map and device** Map name, device ID
External odometry Body velocity, pose, wheel speed Displacement, steering angle, timestamp	**Task and error feedback** Mapping status, mapping progress Service response, error information

9 Frequently Asked Questions

9.1 Deployment, Mapping and Localization

01 How is the software platform accessed?

The browser-based platform supports Ethernet or Wi-Fi connectivity for configuration, mapping, map management and localization monitoring.

02 Is calibration needed without physical odometry?

No. Heading and position calibration establish the relationship between LiDAR and physical odometry and apply only when physical odometry is connected.

03 How large can one mapping session be?

A normal mapping session supports up to 20,000 m² and 2,000 m of cumulative travel. Incremental mapping can extend the map.

04 Can environmental changes affect localization?

Dynamic environments and major structural changes are supported with assistance from LiDAR odometry. Small shifts across large structural areas need attention. Obstruction and prolonged direct sunlight on the camera can also affect localization.

05 Can localization recover after the device is moved?

Coordinate relocalization supports recovery, with position set by dragging on the page or manual entry. Registered, visible QR codes can also assist recovery.

06 Are QR codes mandatory?

Normal operation uses natural environmental features without dedicated QR codes. QR codes can assist localization where features are insufficient.

07 Can devices or floors share a map?

Multiple devices can reuse a map of the same area. Small mounting-height changes require a difference of no more than 50 cm and unchanged surrounding LiDAR features. Different floors use their corresponding maps.

9.2 Data, Integration and Support

01 Are wheel speed or encoders required?

No. V2 Pro can localize independently; physical odometry is optional. Wheel slip may create a mismatch between physical and LiDAR odometry, causing localization drift.

02 Does a communication outage mean localization is lost?

Not necessarily. Localization runs inside the module, so communication loss and localization failure are different states.

03 Does receiving coordinates mean localization is valid?

Not necessarily. Validity also depends on localization status and timestamps. Previously saved positions may still be output when localization is invalid.

04 Is the localization algorithm open source?

The algorithm is proprietary MRDVS technology. Customers integrate control commands and localization results through communication interfaces.

05 Can vehicle positions track pallets or goods?

Goods tracking also requires item identity and loading or unloading records. RFID or barcode information can provide the association.

06 Can other cameras or LiDAR sensors be substituted?

V2 Pro is supplied as an integrated module. Other sensors require project-specific interface and algorithm assessment rather than direct substitution.

07 What technical support is available?

Technical support is available according to project needs. Service scope and fees are agreed by both parties.

10 Global Partners

Regional partners support vehicle localization and system integration needs.

Europe · Japan

Countries

Germany / Japan

Kudan

Visual localization, robot navigation and mapping.

www.kudan.io

Southern Europe

Country

Spain

SleuthX

Real-time localization and item recognition for industry and logistics.

www.sleuthx.com

South America

Country

Brazil

GTP Automation

Warehouse automation, item identification and tracking.

gtpautomation.com

Southeast Asia · India

Country

Singapore

Movel AI

Robotics software, fleet management and system integration.

www.movel.ai

Revision History and Technical Support

Version	Date	Revision
V0.1	2026-10-09	Initial English release: product overview, applications, installation, sensor fusion, performance, operating limits, communications and map use.

Technical Support

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Website: www.mrdvs.com

GitHub: https://github.com/Lanxin-MRDVS

Knowledge Community: https://docs.mrdvs.com/en

LinkedIn: https://www.linkedin.com/company/mrdvs

Scan to visit MRDVS on GitHub